

SUWON, Korea, Republic of

WMO: 471190

Lat: 37.2722N

Lon: 126.9853E

Elev: 38

StdP: 100.87

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.3	-9.3	-21.3	0.6	-7.9	-19.2	0.7	-6.5	6.4	-3.0	5.8	-2.6	1.6	320	0.387

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.7	32.7	25.2	31.4	24.6	30.1	23.7	26.5	30.1	25.9	29.3	25.3	28.5	2.8	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.6	21.0	28.0	25.0	20.1	27.7	24.3	19.4	27.3	83.2	30.1	80.5	29.4	77.9	28.6	32.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
6.1	5.2	4.5	DB	-13.7	34.5	2.3	1.5	-15.4	35.6	-16.8	36.4	-18.1	37.2	-19.8	38.3	(4)
			WB	-14.7	27.5	2.2	1.4	-16.3	28.5	-17.6	29.4	-18.9	30.2	-20.5	31.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	12.9	-2.0	0.5	5.8	12.3	18.0	22.6	25.8	26.5	21.7	15.1	7.5	0.3	
	DBStd	10.41	4.01	3.80	3.60	3.54	2.64	2.04	2.17	2.45	2.59	3.49	4.24	4.12	
	HDD10.0	1197	372	267	137	17	0	0	0	0	0	6	98	301	
	HDD18.3	2738	631	500	388	184	39	1	0	0	4	107	326	559	
	CDD10.0	2256	0	1	7	85	247	379	488	511	352	165	22	0	
	CDD18.3	756	0	0	0	2	28	130	230	252	106	8	0	0	
	CDH23.3	5838	0	0	0	19	187	753	1900	2388	564	27	1	0	
	CDH26.7	1702	0	0	0	2	24	157	578	843	98	1	0	0	
Wind	WSAvg	1.8	1.6	1.8	2.1	2.1	1.9	1.8	1.9	1.9	1.7	1.5	1.6	1.6	
Precipitation	PrecAvg	1296	25	30	47	92	87	119	336	280	148	55	54	23	
	PrecMax	2044	90	116	127	396	245	414	568	851	583	264	151	54	
	PrecMin	850	2	1	5	7	12	5	145	53	3	0	9	1	
	PrecStd	272	22	25	29	86	53	103	140	156	119	61	35	14	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.3	13.9	19.4	25.6	29.1	31.1	34.3	35.0	30.7	25.7	20.1	12.4	
		MCWB	6.4	9.5	11.7	15.5	18.1	20.8	24.9	26.0	23.0	18.0	14.5	8.1	
	2%	DB	7.1	10.6	16.2	23.0	27.0	29.7	32.4	33.3	29.1	23.9	17.7	9.9	
		MCWB	3.5	6.1	9.9	14.4	17.3	20.7	25.0	25.5	21.9	16.8	12.9	6.6	
	5%	DB	5.4	8.5	14.1	21.0	25.4	28.5	31.0	32.1	27.7	22.3	15.8	8.1	
		MCWB	2.3	4.3	8.2	12.6	16.5	20.4	24.8	25.2	20.9	16.2	11.6	5.2	
	10%	DB	3.9	6.5	12.1	19.0	23.8	27.3	29.7	30.8	26.5	20.8	14.1	6.5	
		MCWB	1.3	3.0	7.2	11.9	15.8	19.9	24.2	24.6	20.3	15.4	10.6	3.8	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.8	10.7	13.5	18.3	20.6	24.0	27.2	27.5	25.4	20.8	16.4	9.3
			MCDB	9.3	12.5	17.0	23.2	26.0	26.9	30.4	31.5	28.0	22.4	18.4	10.8
2%		WB	4.5	6.6	11.3	16.0	19.1	22.9	26.4	26.7	24.0	18.7	14.1	7.1	
		MCDB	6.4	9.5	14.8	21.1	23.5	26.1	29.7	30.6	26.7	21.7	16.4	9.3	
5%		WB	3.0	4.9	9.2	14.3	18.2	22.2	25.7	26.2	23.0	17.5	12.5	5.4	
		MCDB	4.8	7.7	12.6	18.8	22.6	25.7	28.9	29.8	25.6	20.7	14.8	7.8	
10%		WB	1.5	3.4	7.6	12.9	17.3	21.4	25.1	25.7	22.0	16.4	11.0	4.0	
		MCDB	3.6	6.1	11.4	17.3	21.9	25.2	28.1	29.1	24.9	19.6	13.5	6.4	
Mean Daily Temperature Range		MDBR	8.1	8.9	9.5	10.5	10.1	8.3	6.3	6.7	8.1	9.7	8.7	8.0	
	5% DB	MCDBR	9.3	11.4	12.7	13.6	12.7	10.4	8.2	8.1	9.5	10.8	10.6	9.6	
		MCWBR	6.4	7.1	6.9	6.1	4.7	3.6	2.9	2.9	3.6	4.7	6.4	6.7	
		MCDBR	8.4	9.9	10.6	11.0	9.8	7.5	6.5	7.0	6.9	8.7	8.7	9.0	
		MCWBR	6.6	7.2	7.0	6.2	5.0	3.4	2.9	3.0	3.5	4.9	6.4	7.0	
Clear-Sky Solar Irradiance	taub	0.440	0.490	0.572	0.613	0.626	0.698	0.662	0.582	0.510	0.499	0.482	0.437		
	taud	1.948	1.859	1.715	1.675	1.697	1.619	1.740	1.930	2.054	1.957	1.944	1.975		
	Ebn at Noon	695	712	693	697	698	648	668	714	744	697	647	664		
	Edh at Noon	140	173	219	240	239	258	227	184	154	154	140	128		
All-Sky Solar Radiation	RadAvg	2.49	3.26	4.28	4.98	5.59	5.21	4.12	4.37	4.12	3.59	2.45	2.18		
	RadStd	0.15	0.26	0.43	0.36	0.52	0.53	0.62	0.62	0.45	0.26	0.25	0.15		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)
(46)	Station Only	N/A	N/A	N/A	N/A	+0.49	+0.53	N/A	N/A	+70	N/A	N/A
(47)	Regional (46 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+94	+50	N/A

Nomenclature: See separate page